Energy and Efficiency Worksheet

For the following devices state what type of Energy is used from this list:

Chemical Kinetic Heat (thermal) Sound Electrical Light

Light  bulb: Input Energy _____

Useful Energy _____ Waste Energy _____

Electric  Drill: Input Energy _____

Useful Energy _____

Waste Energy_____ & _____



TV: Input Energy _____

Useful Energy _____ & _____

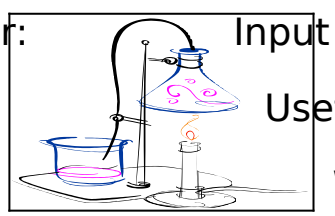
Waste Energy _____

Car 

engine: Input Energy _____

Useful Energy _____

Waste Energy _____ & _____

Bunsen burner: 

Input Energy _____

Useful Energy _____

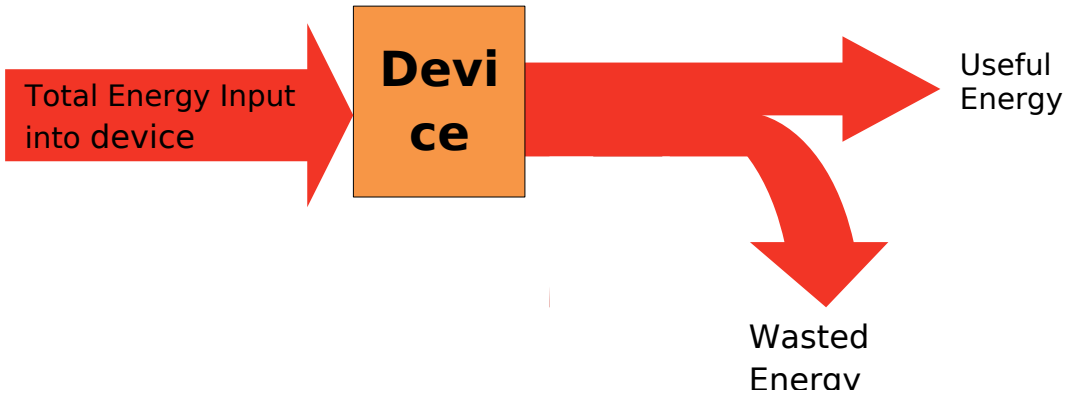
Waste Energy_____ & _____

A Sankey diagram represents the energy transfer through a device. Knowing that energy **cannot be created or destroyed**, energy input must equal the total energy output:

$$Total\ input\ energy = useful\ energy\ delivered + energy\ wasted$$

A Sankey diagram shows this:

Note: the **width** of the arrows demonstrates the **amount** of energy.



Draw your own Sankey diagram for a 60W filament bulb given that:

Total input energy =60W
Useful (light) energy = 10W
Waste (heat) energy = 50W

Use the squares below as a guide for drawing your arrows.

The greater the percentage of energy that is transformed into useful energy in a device, the more efficient a device is:

$$Efficiency\ of\ Device = \frac{Useful\ energy\ transferrered\ by\ the\ device\ (output)}{Total\ energy\ supplied\ ¿} the\ device\ (input)\ ¿$$

Efficiency can be written as either a number (which is never more than 1) or as a percentage (never more than 100%). Neither have units.

For example: A light bulb radiates 10W of energy as light, for every 60W of electrical energy we supply to it.

$$Efficiency\ of\ a\ lightbulb = \frac{10}{60} = 0.17\ (as\ a\ number)$$

$$Efficiency\ of\ a\ lightbulb = \frac{10}{60}\ x\ 100 = 17\%\ (as\ a\ percentage)$$

Now **complete** the table below, **calculating the efficiency** as both a number and a percentage:

	Energy in	Energy out	Efficiency?
Electric Drill	160W	Kinetic 90W Sound 30W Heat 40W	
Hair drier	180W	Heat 170W Sound 10W	
Mobile phone charger	1W	0.8W Electrical 0.2W Heat	
Electric hob	1500W	Heat 1300W Light 150W Sound 50W	
Kettle	2.5kW	Heat 2.2kW Sound 0.3kW	

Extension: Using the equation:

$$Efficiency\ of\ Device = \frac{Useful\ energy\ transferrered\ by\ the\ device\ (output)}{Total\ energy\ supplied\ ¿} the\ device\ (input)\ ¿$$

Fill in the gaps in the table below.

	Energy in	Energy out	Efficiency?
Food mixer	120W	Useful energy 80W Waste energy 40W	
TV	180W	Useful Energy W Waste energy 36W	0.8 80%
Ipod	W	Useful Energy 0.6W Waste energy 0.4W	
Car engine	30kW	Useful Energy kW Waste energy kW	0.45 45%
X box	260W	Useful Energy W Waste energy 70W	
Mobile phone	W	Useful Energy 3.6W Waste energy W	0.45 45%